

CHAPTER 1

INTRODUCTION

Water bodies such as lakes, ponds, and rivers play a vital role in maintaining ecological balance, supporting biodiversity, and regulating urban water systems. They act as natural reservoirs that store rainwater, recharge groundwater, and help in flood control. However, rapid urbanization, population growth, and weak enforcement of environmental regulations have led to increasing encroachments on these water bodies. These encroachments reduce storage capacity, disrupt natural drainage systems, and significantly increase the risk of urban flooding and environmental degradation.

In many regions, grievance redressal systems have been implemented to allow citizens to report civic issues, including illegal encroachments. However, these systems often lack proper location-based verification, transparency, and time-bound action mechanisms. Complaints may be delayed, misrouted, or ignored, resulting in ineffective resolution and reduced public trust in governance systems.

To address these challenges, this project proposes the development of NeerRaksha a Geo-Verified, Time-Bound Civic Escalation System that enables citizens to report water body encroachments using geo-tagged data. The system ensures that complaints are validated based on location, processed within a predefined time frame, and escalated automatically if not resolved. Additionally, the system incorporates an AI-based module to analyze complaint descriptions and classify their severity levels, enabling priority-based allocation and faster response from authorities. The name NeerRaksha is derived from the Sanskrit words for water

and protection, encapsulating the system's core mission of protecting water resources through technology-enabled civic intelligence.

India alone has documented the disappearance or severe degradation of thousands of water bodies over the past three decades. Studies from states such as Tamil Nadu, Karnataka, Telangana, and Maharashtra indicate that urban lakes have shrunk by thirty to sixty percent of their original extent due to illegal constructions, unauthorized land filling, and encroachment by industrial and commercial establishments. This alarming trend has direct consequences for urban resilience, as the capacity of cities to absorb and manage monsoon rainfall depends critically on the integrity of their natural drainage and storage systems. The administrative systems currently in place are inadequate to this challenge — most state and municipal governments operate complaint portals that accept grievances but lack the infrastructure to verify authenticity, prioritize by severity, or enforce timely resolution.

NeerRaksha is conceived as a comprehensive response to this governance gap. It integrates geospatial validation, artificial intelligence, and automated administrative workflows into a single unified platform that transforms water body complaint management from a passive, paper-based process into an active, data-driven governance framework. The system serves two equally important communities: citizens who need a simple, trustworthy channel to report encroachments, and government administrators who need accurate, prioritized, and actionable complaint data supported by performance analytics. This chapter provides the foundational framing for the system. Section 1.1 articulates the specific problem. Section 1.2 provides a system overview. Section 1.2.1 describes key functional components. Section 1.3 outlines assumptions and dependencies.

1.1 Problem Definition

Encroachments on water bodies have become a serious environmental and urban planning issue due to unregulated development and lack of strict enforcement. These encroachments include illegal constructions, land filling, waste dumping, and unauthorized use of lakebeds or riverbanks. Over time, such activities reduce the natural capacity of water bodies, leading to flooding, habitat destruction, and depletion of groundwater recharge zones.

Although existing complaint systems allow citizens to report such issues, they suffer from several critical limitations. There is no reliable mechanism to verify whether a complaint actually falls within a water body boundary. There is no strict enforcement of response timelines, and authorities are not held accountable for delays. Complaints are often not prioritized by severity, meaning critical encroachments may be queued behind trivial ones. As a result, complaints remain unresolved for long periods, reducing the effectiveness of these systems and eroding public trust in environmental governance.

This project addresses these challenges by developing a system that integrates geo-spatial validation, time-bound response mechanisms, and automated escalation. By ensuring that complaints are verified, prioritized, and handled within defined timelines, the system improves efficiency, transparency, and accountability in managing water body encroachments. The multi-dimensional nature of the encroachment problem requires a solution that operates at the technical level accurate geo-fencing, severity classification, and routing at the governance level enforceable deadlines and hierarchical accountability and at the social level accessible interfaces that encourage widespread citizen participation.

A particularly critical dimension of the problem is temporal. Encroachments typically begin as minor, easily reversible activities such as small dumping grounds or temporary structures, and escalate over time into permanent, difficult-to-remove constructions. A system that enables early detection and rapid response can interrupt

this progression before it reaches an irreversible stage. The legal and jurisdictional complexity adds further difficulty, as water bodies may fall under the jurisdiction of multiple agencies including municipal corporations, irrigation departments, lake development authorities, and pollution control boards. The system must accommodate this complexity while maintaining simplicity that encourages citizen participation.

1.2 System Overview

The proposed system is a web-based platform designed to monitor, validate, and manage complaints related to water body encroachments. It integrates citizen participation with geospatial technologies and administrative workflows to create an effective governance solution. Users can submit complaints along with geo-tagged photographs and location coordinates. The system validates these coordinates using geo-fencing techniques by comparing them with official water body boundary datasets. Once validated, the complaint is forwarded to the appropriate authority and a Service Level Agreement (SLA) deadline is assigned.

The system continuously tracks the status of each complaint and displays it using a color-coded indicator based on the SLA timeline. If the complaint is not resolved within the specified time, it is automatically escalated to higher authorities. A public dashboard displays complaint statistics, response times, and authority performance, ensuring transparency and accountability. The operational workflow is a five-stage sequential pipeline: complaint submission with geo-tagged data; geo-validation against official boundary datasets; AI-based risk assessment and severity classification; SLA assignment and authority routing; and continuous monitoring with automated escalation and public reporting.

1.2.1 Key Components

The NeerRaksha system comprises six primary functional components, each designed as a self-contained module with well-defined interfaces that support independent development, testing, and enhancement.

Complaint Submission Module: Allows citizens to report encroachments by submitting complaints along with geo-tagged images, location coordinates, and descriptions. It captures essential information such as user details, time, and location of the issue. The interface integrates an interactive map powered by Leaflet.js that displays water body boundaries and the user's GPS position, enabling informed and accurate complaint submissions.

Geo-Validation Module: Verifies whether the reported location falls within the boundary of a water body. It uses geospatial datasets such as GeoJSON or shapefiles and performs spatial analysis using a ray-casting point-in-polygon algorithm to validate the authenticity of the complaint. This critical module ensures that only geographically relevant complaints enter the processing pipeline.

Data Management Module: Responsible for storing and managing all system data, including complaints, user details, authority information, and water body boundary datasets. It ensures efficient data handling and retrieval for all system operations, implementing appropriate concurrency controls to prevent data corruption under simultaneous access.

SLA and Escalation Engine: Assigns a predefined deadline to each verified complaint and monitors its progress continuously through scheduled background tasks. If the complaint is not resolved within the given time, the system automatically escalates it to higher authorities, ensuring time-bound action and hierarchical accountability.

Visualization and Dashboard Module: Provides an interactive interface for users and administrators to track complaint status, view trends, and analyze

performance metrics. It includes graphical representations using Chart.js and maps using Leaflet.js for better spatial understanding and administrative decision support.

Real-Time Monitoring for Water Body Protection: Provides real-time updates on complaint status and enables proactive monitoring of encroachments. By combining citizen participation with automated processes, it supports efficient protection of water bodies and provides data-driven insights for environmental governance.

1.3 Assumptions and Dependencies

1.3.1 Assumptions

The system assumes that the geo-location data captured from user devices is sufficiently accurate for validation. Modern GPS-enabled smartphones typically provide location accuracy within five to fifteen meters under clear sky conditions, which is adequate for the water body boundary validation performed by the system. A configurable buffer zone around water body boundaries accommodates the GPS accuracy limitations of typical consumer devices. It also assumes that water body boundary datasets obtained from reliable government sources such as the National Water Informatics Centre and state irrigation departments are correct and up to date. The predefined SLA durations are assumed to be appropriate for effective complaint resolution, based on established e-governance best practices and regulatory guidelines.

It is further assumed that authorities will respond to escalated complaints and that automated escalation mechanisms will improve accountability by creating visible performance records. Citizen participation is also assumed to be genuine and active, with users submitting reports based on actual observations rather than fabricated information. While the geo-validation mechanism provides a layer of

automated authenticity checking, the system relies on good-faith participation for its broader effectiveness.

1.3.2 Dependencies

The system depends on the availability of accurate and updated geospatial data for water body boundaries. It also requires reliable internet connectivity for real-time complaint submission and tracking. Integration with mapping services such as OpenStreetMap is necessary for location visualization and base map rendering. Additionally, backend processing requires sufficient computational resources for handling geo-validation and data management tasks. The Node.js runtime environment with Express.js and supporting npm packages must be properly configured on the server. The effectiveness of the system also depends on cooperation from government authorities and their willingness to engage with the system's administrative interface for complaint handling and escalation. Without active participation from designated authorities, the technical excellence of the system cannot translate into improved environmental outcomes.

The urgency of addressing water body encroachments is underscored by multiple compounding factors. Climate change is intensifying the monsoon cycle, increasing the frequency and severity of extreme rainfall events. Cities that have allowed their natural water bodies to shrink are experiencing progressively worse flooding during high-rain seasons, as the reduced buffer capacity of lakes and wetlands translates directly into increased surface runoff volumes. Simultaneously, rapid population growth continues to create pressure on urban land markets, making encroachment economically attractive to developers who face little consequence under the present enforcement regime. NeerRaksha intervenes precisely at this nexus of environmental urgency and governance failure.

The social dimensions of water body encroachment are equally significant. Urban water bodies have historically served as centres of community life, providing livelihoods for fishing communities, cultural sites for local festivals, and accessible

public green space in otherwise densely built neighbourhoods. Their loss is not merely an environmental statistic but a tangible diminishment of community heritage and social equity. Low-income communities living near urban water bodies are disproportionately affected by both the encroachment itself which deprives them of livelihood resources and by the flooding consequences that follow. A responsive civic reporting system that enables these communities to have their observations heard and acted upon represents a form of environmental justice as much as technical governance innovation.

The governance landscape in which NeerRaksha operates is characterised by institutional fragmentation. Water bodies in Indian cities may simultaneously fall under the jurisdictions of the Urban Local Body, the State Irrigation Department, the Lake Development Authority, the Revenue Department, and the State Pollution Control Board. Different aspects of encroachment structural construction, waste dumping, drainage modification, and commercial exploitation may each trigger the mandate of a different agency. This jurisdictional complexity has historically been exploited by encroachers who rely on inter-agency confusion to avoid enforcement action. NeerRaksha addresses this by providing a unified complaint record that can be routed to multiple authorities simultaneously and by maintaining a transparent escalation trail that makes jurisdictional hand-off failures visible.

1.4 Scope and Limitations

The current implementation of NeerRaksha is designed and tested for urban water bodies in India, with GeoJSON boundary datasets sourced from the National Water Informatics Centre and OpenStreetMap. The system is scoped to handle five encroachment categories: illegal construction, land filling, waste dumping, drainage obstruction, and commercial encroachment. These categories cover the most commonly reported and environmentally damaging forms of water body degradation

observed in Indian cities. The system is not designed to handle ocean or tidal water boundaries, trans-boundary river disputes, or encroachments that occur underwater or below sediment level, as these require specialised monitoring technologies beyond citizen-reported observations.

The current implementation stores data in JSON flat files, which is adequate for the demonstration and pilot deployment scale envisaged in this project. The system is designed with a clear data access abstraction that supports migration to a relational or document database without requiring changes to the business logic modules. Similarly, the AI classification module uses a keyword lexicon approach that provides reliable performance for the English-language complaint descriptions that dominate current test data, but may require augmentation with regional language models for deployment in areas where citizens are more comfortable reporting in Tamil, Hindi, Telugu, or other scheduled languages of India.

The geo-validation accuracy of the system depends on the completeness and currency of the water body boundary dataset. Areas where official GIS datasets are absent or significantly outdated will yield reduced validation performance. The system acknowledges this limitation by routing all boundary-edge cases to human review rather than automatic rejection, ensuring that no genuine encroachment complaint is lost due to dataset gaps. Future work to integrate regularly updated satellite-derived water body extents will address this limitation progressively.

1.5 Significance of the Work

The NeerRaksha system makes several contributions that distinguish it from prior work in civic technology and environmental monitoring. First, it is the only system that combines automated geo-validation, AI-based severity classification, and time-bound escalation in a single integrated platform specifically designed for water body encroachment management. Second, it demonstrates that the ray-casting

point-in-polygon algorithm, previously applied primarily in desktop GIS software, can be implemented effectively in a server-side JavaScript environment with sub-second response times suitable for real-time web applications. Third, it establishes a replicable architecture that can be adapted to other categories of environmental encroachment including forest boundary violations, coastal regulatory zone encroachments, and industrial pollution near ecologically sensitive areas.

From a governance perspective, the system demonstrates that automated accountability mechanisms specifically the combination of binding SLA deadlines and automatic escalation with public visibility can be designed into civic systems at relatively low cost and without requiring changes to existing legal frameworks. The system works within the existing administrative hierarchy rather than replacing it, adding a technological layer that makes the hierarchy's performance visible and accountable without requiring institutional restructuring. This pragmatic design philosophy is likely to facilitate adoption by government agencies that are cautious about systems that challenge existing authority structures.

The broader contribution of this work to smart city research is the demonstration that effective environmental governance solutions require co-design of the citizen-facing and authority-facing interfaces, with equal attention to the usability needs of both communities. Smart city projects that optimise for data collection without providing actionable administrative workflows fail to achieve impact. NeerRaksha's design integrates the complete feedback loop from citizen observation through administrative action to public performance reporting, closing the accountability gap that undermines the effectiveness of most existing civic reporting platforms.

The potential policy impact of NeerRaksha extends beyond the immediate application to water body protection. The system's architecture demonstrates a generalizable pattern for technology-enabled environmental governance that could be adapted to other domains where spatial authenticity verification, severity-based

prioritization, and time-bound accountability are needed. Coastal regulatory zone violations, forest boundary encroachments, and protected area buffer zone violations all share the same structural characteristics as water body encroachments and could benefit from the same integrated approach. By documenting the design rationale, implementation choices, and performance results of NeerRaksha in detail, this project provides a replicable blueprint for future civic technology initiatives in the environmental governance space, contributing to the growing body of evidence that technology-enabled participatory governance can produce measurable improvements in environmental outcomes.

CHAPTER 2

LITERATURE SURVEY

The increasing threat to urban water bodies due to encroachments has drawn the attention of researchers across disciplines including geospatial science, artificial intelligence, urban governance, and civic technology. Several studies have proposed or evaluated platforms and techniques relevant to the problem addressed by NeerRaksha. This section reviews the most significant related works, critically assessing their contributions and limitations, and positioning the proposed system within the existing research landscape.

Meier [1] presented Ushahidi, a pioneering crowdsourced data collection platform originally developed during the 2007 Kenyan post-election crisis. The system enables citizens to submit real-time reports through mobile devices, which are visualized on an interactive geographic map, integrating data from SMS, web applications, and social media. Ushahidi demonstrated that ordinary citizens with basic mobile devices could contribute meaningful intelligence to crisis management operations, achieving deployment in over 150 countries. However, it primarily focuses on data collection and visualization and lacks mechanisms for validating report authenticity, enforcing time-bound resolution workflows, or applying hierarchical escalation. The absence of geo-validation against official boundary datasets and the lack of any severity classification mechanism make Ushahidi unsuitable for the specialized regulatory context of water body encroachment management, where administrative accountability and geographic precision are critical requirements.

Steinberg [2] developed FixMyStreet, a civic engagement platform allowing citizens to report local infrastructure issues such as potholes and waste management

problems using location data. The platform automatically routes complaints to appropriate authorities based on geography and has been successfully deployed across the United Kingdom and multiple other countries. While FixMyStreet established important design principles including map-based submission and public complaint visibility, it lacks geographic validation of complaints against official boundary datasets, severity-based prioritization, and strict enforcement of resolution timelines. There is no automated escalation when deadlines are missed, and no AI-based assessment of complaint urgency, making it insufficient for specialized environmental regulatory applications.

Bloom [3] proposed Open311, an open standard designed to streamline communication between citizens and municipal services through standardized APIs. It was adopted by cities including San Francisco, Washington D.C., and Chicago, enabling interoperability between diverse civic complaint platforms. However, Open311 functions as an integration standard rather than an intelligent complaint management system. It does not include geo-validation, automated escalation, or AI-based prioritization mechanisms, and the quality of complaint resolution remains entirely dependent on the responsiveness of individual city departments.

Longley et al. [4] provided a comprehensive treatment of Geographic Information Systems (GIS) and spatial data analysis techniques. Of particular relevance is the detailed explanation of the ray-casting point-in-polygon algorithm, which determines whether a geographic coordinate falls within a polygon boundary by counting ray-edge intersections. The text also covers GeoJSON and shapefile formats for boundary representation and bounding-box pre-filtering for efficient spatial queries. While GIS offers powerful analytical capabilities, this work does not address citizen complaint management, administrative accountability, or time-bound governance workflows the precise gap that NeerRaksha occupies.

Liu [5] presented a foundational text on Sentiment Analysis and Opinion Mining, covering Natural Language Processing techniques for extracting sentiment

and intent from textual data. The keyword-based severity lexicon approach and token-level classification methodology described in this work directly informed the design of NeerRaksha's AI-based risk classification module. The text established that domain-specific lexicons combined with weighted scoring mechanisms can achieve reliable classification accuracy for short, structured texts such as civic complaint descriptions.

Iqbal and Iqbal [6] conducted a study on land use detection using remote sensing and GIS techniques in the Rawalpindi Division of Pakistan. Using multi-temporal Landsat imagery and supervised maximum likelihood classification, they documented significant reductions in water body area over a 25-year study period due to urbanization and agricultural encroachment. While the study demonstrates the power of satellite imagery for detecting land use changes at macro scale, it does not provide mechanisms for citizen participation, real-time reporting, or administrative complaint resolution. Its findings reinforce the scientific urgency of the NeerRaksha mission by quantifying the scale of water body degradation.

Singh and Rai [7] examined wetland changes using remote sensing and GIS, applying NDWI (Normalized Difference Water Index) and image differencing to detect temporal variations in wetland extent and health. The study identified agricultural encroachment, urban expansion, and drainage modification as primary causes of wetland degradation and recommended the development of spatiotemporal monitoring databases. The recommendation for systematic continuous monitoring directly informed the design of NeerRaksha's complaint database and dashboard, while the macro-scale remote sensing approach complements the micro-scale citizen reporting mechanism of the proposed system.

Song et al. [8] provided a comprehensive overview of Smart Cities, examining how IoT sensors, big data analytics, cloud computing, and AI can be integrated to improve urban governance across transportation, energy, healthcare, and environmental domains. The authors argue convincingly that effective smart city

solutions must transcend technological innovation to integrate institutional structures and citizen participation. This principle directly influenced the design philosophy of NeerRaksha as a governance innovation enabled by technology, and the emphasis on citizen participation as a core smart city component reinforced the decision to center the system's design around a citizen-friendly reporting interface.

Jain and Sharma [9] studied urban flood management using GIS and remote sensing, establishing a quantitative link between water body encroachment and the amplification of peak flood discharge. Their analysis of Digital Elevation Models and historical flood data demonstrated that even partial reduction in water body area significantly increases downstream flood risk. The study argues for preserving natural water bodies as a cost-effective flood mitigation strategy. This work provides critical quantitative justification for the NeerRaksha mission by establishing the direct link between prompt encroachment response and urban flood resilience.

Based on the above review, it is evident that while several platforms address civic complaint management and environmental monitoring in isolation, no existing work integrates automated geo-validation, AI-based risk prioritization, time-bound SLA enforcement, and transparent public accountability within a single system specifically designed for water body protection. Table 2.1 presents a comparative analysis of the reviewed works against these key criteria.

System / Study	Geo-Validation	AI Classification	SLA Enforcement	Public Dashboard
Ushahidi [1]	No	No	No	Basic
FixMyStreet [2]	Partial	No	No	Yes
Open311 [3]	No	No	No	No
GIS & Science [4]	Analytical	No	No	No
NLP/Sentiment [5]	No	Yes	No	No

Land Use [6]	Analytical	No	No	No
Wetland Study [7]	Analytical	No	No	No
Smart Cities [8]	No	No	Partial	Partial
Urban Flood [9]	Analytical	No	No	No
NeerRaksha	Yes (automated)	Yes (NLP)	Yes (automated)	Yes (real-time)

Table 2.1: Comparative Analysis of Related Works

CHAPTER 3

PROPOSED SYSTEM

This chapter presents the detailed design and architecture of the NeerRaksha system. It describes each module's function, implementation approach, and interaction with other components, providing the technical foundation for the implementation work described in Chapter 4.

3.1 System Architecture

The proposed system is a Geo-Verified, Time-Bound Civic Escalation System designed to manage and resolve complaints related to water body encroachments. The system integrates web technologies, geospatial validation, artificial intelligence, and automated workflow mechanisms to ensure efficient complaint handling.

The architecture consists of five major layers working in concert to process citizen complaints from submission through resolution. The User Interface Layer provides the citizen-facing web application built on HTML5, CSS3, and JavaScript with Leaflet.js for interactive mapping. The Application Layer implements server-side business logic using Node.js and Express.js, coordinating all processing modules. The Geo-Validation Layer performs spatial containment analysis against GeoJSON water body boundary datasets using the ray-casting algorithm. The AI-Based Risk Assessment Layer analyzes complaint descriptions using Natural Language Processing to classify severity levels. The Database Layer manages persistent storage of all complaint, user, authority, and boundary data.

These layers communicate through well-defined internal interfaces, maintaining the separation of concerns that enables each component to be independently developed, tested, and enhanced. The layered architecture also supports future scaling, as individual layers can be upgraded or replaced without disrupting the overall system. The application follows RESTful API conventions for communication between the frontend and backend, enabling potential integration with external systems including official government portals and mobile applications.

A Scheduled Task Engine operates independently as a background process that periodically examines all active complaints, computes their elapsed time against SLA deadlines, and triggers escalation actions for overdue cases. This engine is critical to the system's accountability mechanism, ensuring that unresolved cases are automatically escalated rather than requiring manual administrative oversight.

The overall system workflow begins when a citizen submits a complaint through the web application. The complaint passes through the geo-validation layer, which confirms its spatial authenticity. It then passes through the AI classification layer, which assigns a risk level. The SLA engine assigns a resolution deadline and routes the complaint to the appropriate authority. The background task engine monitors progress and triggers escalations as needed. Throughout this process, all status updates are visible on the public dashboard, ensuring transparency for both citizens and administrators.

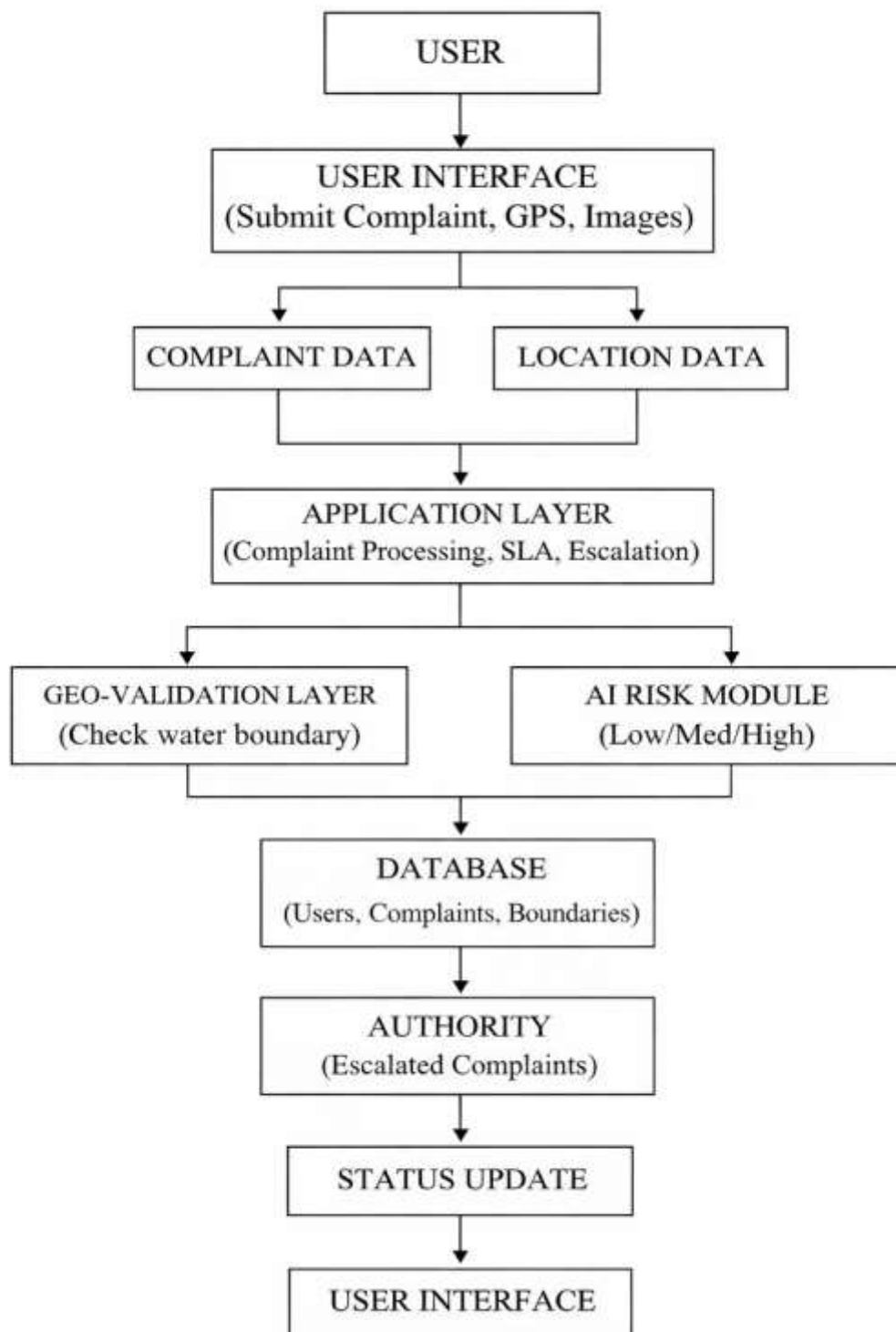


Fig. 3.1: System Architecture of NeerRaksha

3.2 Module Explanation

3.2.1 Complaint Submission and Data Processing

In this module, users submit complaints related to water body encroachments through the web application. Each complaint includes details such as description, images, location coordinates (latitude and longitude), and timestamp. The submission interface integrates an interactive Leaflet.js map that displays official water body boundaries overlaid on OpenStreetMap base tiles, providing citizens with visual spatial context that helps them understand whether their observation falls within a recognized water body area.

The system processes the submitted data through the following sequential steps. First, client-side validation ensures all required fields are populated and submitted images meet size and format requirements. Second, the form data is transmitted to the backend as a multipart HTTP POST request. Third, the backend extracts location coordinates and stores complaint details in the database with a unique identifier and timestamp. Fourth, the data is prepared for geo-validation and further processing by the subsequent modules.

This module ensures that all complaints are captured accurately and structured properly for validation and analysis. The unique complaint identifier generated at this stage is communicated to the submitting citizen as their reference number for tracking, creating a persistent link between the citizen and their report throughout its lifecycle in the system. The module also implements security measures including input sanitization to prevent injection attacks, file type and size validation to prevent malicious file uploads, and session management to maintain user authentication state.

3.2.2 Geo-Validation Module

The Geo-Validation Module is responsible for verifying the authenticity of the complaint location. It uses geospatial datasets such as GeoJSON or shapefiles that contain official water body boundaries sourced from government GIS portals and authoritative open geospatial data repositories.

The module performs spatial analysis to check whether the submitted coordinates fall within the defined boundary of a water body. This is achieved using the ray-casting point-in-polygon algorithm, which determines containment by counting the intersections of a horizontal ray from the complaint point with the polygon boundary edges. An odd intersection count indicates the point is inside the polygon; an even count indicates it is outside. If the complaint is valid, it is forwarded to the AI-based risk assessment stage. If not, it is flagged for human review rather than automatically discarded, preserving the possibility that genuine complaints near boundary edges might require manual consideration.

The module handles edge cases including points on polygon boundaries, MultiPolygon geometries representing complex water bodies, and GPS accuracy limitations. A configurable buffer zone extending the effective boundary accommodates positioning errors of up to fifteen meters. A bounding box spatial index pre-filters candidate polygons before the full ray-casting test is applied, ensuring sub-second validation response times even for datasets covering hundreds of water bodies.

3.2.3 AI-Based Risk Assessment Module

This module uses Artificial Intelligence to analyse the complaint description and determine the severity of the reported issue. Natural Language Processing (NLP) techniques are used to identify keywords, patterns, and context within the complaint text submitted by the citizen.

Based on the analysis, the complaint is classified into one of three risk levels. Low Risk corresponds to minor issues with minimal immediate impact, such as small debris accumulation or temporary minor obstructions. Medium Risk applies to moderate issues that may worsen over time, such as regular waste dumping or partial boundary encroachment. High Risk identifies critical issues requiring immediate attention, such as permanent illegal construction, large-scale land filling, or complete blockage of inflows.

The classification algorithm operates through a multi-step process. The complaint description is tokenized and cleaned by removing common stop words. The remaining tokens are matched against a domain-specific severity lexicon that maps encroachment-related terms to severity scores. A category weight is applied based on the selected encroachment type. The final composite score determines the risk classification. High-risk complaints receive 24-hour SLA deadlines, Medium-risk complaints receive 72-hour deadlines, and Low-risk complaints receive 7-day deadlines.

3.2.4 SLA Management and Escalation Module

Once a complaint is validated and classified, a Service Level Agreement (SLA) is assigned based on its severity level. The SLA defines the time frame within which the complaint must be resolved by the assigned authority. This time-bound framework creates the structural accountability that distinguishes NeerRaksha from conventional complaint portals where deadlines are advisory rather than enforced.

The system continuously monitors complaint status through a scheduled background task that executes at regular intervals. The monitoring produces a color-coded classification: Green signifies the complaint is within its SLA window and being actively processed; Yellow warns that remaining resolution time has fallen below 25 percent; Red appears when the deadline has been crossed. When Red status

is detected, the escalation mechanism is activated, proceeding hierarchically from field officer (Level 0) to ward-level supervisor (Level 1) to district environmental officer (Level 2) to state-level authority (Level 3). Each escalation event generates notifications to the newly assigned authority and the complainant, and all escalation events are logged with timestamps creating an immutable audit trail.

The SLA and escalation parameters are configurable by system administrators to accommodate regional variations in administrative capacity and regulatory requirements. This configurability ensures that the system can be deployed across different administrative contexts without requiring code changes.

3.2.5 Authority Allocation Module

This module is responsible for assigning each validated complaint to the most appropriate government authority based on both geographic location and severity level. It plays a crucial role in ensuring that complaints are handled efficiently, without unnecessary delays or misrouting, which are common issues in traditional grievance systems.

The module operates using a rule-based allocation mechanism that considers the complaint's geo-validated location, assigned risk level, and category of encroachment. A key component is the authority registry, implemented as a configurable JSON data structure that maintains mappings between geographic zones (wards, municipalities, districts), complaint categories, and designated government officials. Each entry includes authority ID, designation, jurisdiction boundary, escalation level, and contact details.

The allocation logic follows a hierarchical governance model. Lower-risk or routine issues are assigned to local authorities such as ward-level officers or field inspectors. High-risk complaints involving permanent constructions, large-scale land filling, or critical environmental impact are directly routed to district

environmental officers or regulatory bodies. The module also incorporates a load-aware assignment strategy to prevent overloading a single authority, distributing complaints among multiple officials within the same jurisdiction. The system maintains a complete log of all authority assignments, including timestamps, authority details, and any reassignment or escalation events, establishing a transparent chain of responsibility essential for accountability.

3.2.6 Web Application and Visualization Module

A user-friendly web application is developed to provide interaction between users and the system. The application serves two distinct user groups — citizens submitting and tracking complaints, and administrators managing the complaint resolution process — with interfaces tailored to the specific needs and technical literacy levels of each group.

The citizen interface includes a home page presenting system information and recent statistics, a complaint submission page with integrated Leaflet.js map and data entry form, and a complaint tracking page where citizens can monitor their report status using their unique complaint identifier. Status updates are displayed with color-coded indicators showing whether the complaint is pending, in progress, or resolved, with the SLA deadline and time remaining clearly visible.

The administrative dashboard provides a comprehensive operational view including real-time complaint counts by status and risk level, an interactive map showing complaint hotspots overlaid on water body boundaries, a bar chart of monthly complaint volumes using Chart.js, pie charts showing distribution by category and authority, a sortable and filterable complaint list with individual record access, and authority performance metrics. These features enable data-driven management of the complaint resolution process and support strategic decision-making about resource allocation and enforcement priorities.

3.3 System Security Architecture

Security is a critical design consideration for NeerRaksha, as the system handles personally identifiable information submitted by citizens reporting sensitive environmental violations, and administrative interfaces that control the complaint resolution process. The security architecture is designed around the principle of defence in depth, implementing multiple independent security controls at different layers of the system so that a failure at any single layer does not compromise overall system integrity.

Authentication and authorisation are implemented through a session-based mechanism using the express-session middleware. Citizen users are not required to create accounts, reducing friction in the complaint submission process and encouraging participation from citizens who distrust data collection by government-affiliated systems. However, citizens receive a unique complaint identifier at submission that serves as a single-factor authentication token for accessing their specific complaint status without exposing the full complaint database. Administrative users authenticate with username and password credentials stored as hashed values using the bcrypt key derivation function, which protects stored credentials against offline dictionary attacks even in the event of database disclosure.

Input validation and sanitization are applied at both the client and server layers. Client-side validation provides immediate user feedback and reduces unnecessary server load, while server-side validation provides the authoritative security check that cannot be bypassed by client-side script modification. All text inputs are sanitized to remove potentially dangerous HTML and script content before storage, preventing stored cross-site scripting attacks. File uploads are validated for MIME type and file extension, and uploaded images are processed through a size-normalization step that strips EXIF metadata before storage, preventing the disclosure of precise GPS coordinates embedded in smartphone photographs that could identify the exact location and device of the complainant.

API rate limiting is applied at the complaint submission endpoint to prevent automated abuse. A sliding window rate limiter allows a maximum of ten complaint submissions per hour per IP address, which is adequate for legitimate citizen use while preventing automated flooding of the complaint database. Administrative API endpoints are protected by session authentication and additionally by CSRF tokens to prevent cross-site request forgery attacks that could allow malicious websites to perform administrative actions on behalf of authenticated administrators.

3.4 Performance Design Considerations

The performance requirements of NeerRaksha are driven by two distinct usage patterns: peak submission periods during or immediately after significant rainfall events when large numbers of citizens may simultaneously submit encroachment reports, and continuous background monitoring by the SLA engine that must efficiently process potentially hundreds of active complaints. The system is designed to handle both patterns without degradation of response time or data integrity.

The geo-validation module is the most computationally intensive operation in the complaint processing pipeline. A naive implementation that tests each submitted coordinate against all water body polygons in the dataset would scale poorly as the dataset grows. The implemented optimization uses a bounding box spatial index that pre-filters candidate polygons before the full ray-casting test, reducing the average number of full polygon tests per validation from the full dataset size to typically two or three. This optimization achieves sub-second validation response times for datasets covering up to five hundred water bodies, which is adequate for city-level or district-level deployments. For regional or national deployment, a more sophisticated spatial indexing structure such as an R-tree would be required.

The JSON flat file storage approach provides adequate performance for the demonstration scale of the current implementation but introduces a sequential write bottleneck under high concurrent load. The write queue implemented in the data

access layer serializes concurrent write operations, preventing data corruption at the cost of reduced write throughput. For production deployment at city scale, migration to a transactional database system such as PostgreSQL with PostGIS spatial extensions would eliminate this bottleneck while adding native geospatial query capabilities that would further optimize the geo-validation process.

The dashboard data retrieval is optimized through pre-computation of aggregate statistics at the time of complaint status changes rather than at query time. When an administrator views the dashboard, the system retrieves pre-computed metrics from the metrics store rather than executing aggregate queries over the full complaint database. This approach trades write latency for read performance, which is appropriate for a dashboard that is read frequently by multiple administrators but updated only when complaint statuses change.

3.5 Deployment Architecture

NeerRaksha is designed for deployment on a standard Linux server environment, with no dependency on proprietary cloud services or specialized hardware. The minimal deployment configuration requires a server with two CPU cores, four gigabytes of RAM, and fifty gigabytes of storage, which is within the hardware specification of typical government server infrastructure. The Node.js runtime and all required npm packages are installed once during initial deployment, and the application is managed by the PM2 process manager which provides automatic restart on failure, log rotation, and process monitoring.

The recommended production deployment configuration uses a reverse proxy such as Nginx in front of the Node.js application server. The Nginx layer handles SSL/TLS termination, serving static frontend assets directly without involving the application server, and implementing additional rate limiting and request filtering. This configuration is standard for Node.js web applications and is well-documented

in the DevOps community, reducing the deployment support burden on government IT departments.

The GeoJSON water body boundary data is loaded into memory at application startup and maintained as an in-memory spatial index throughout the server's lifecycle. This design prioritizes validation response time by eliminating disk I/O from the critical path of complaint processing. The boundary data file is configurable and can be updated by replacing the data file and restarting the application, which typically completes within ten seconds. More sophisticated update strategies, including hot reload of boundary data without application restart, are planned for future enhancement.

For a multi-city or state-wide deployment, the architecture supports horizontal scaling through the deployment of multiple application server instances behind a load balancer. The only state that must be shared between instances is the session store, which would be migrated to a Redis-based shared session store in a horizontally scaled deployment. The complaint data store would be migrated to a shared database server accessible to all application instances. This scaling path is straightforward and does not require architectural redesign of the application itself.

CHAPTER 4

IMPLEMENTATION AND RESULTS

This chapter describes the complete implementation of the NeerRaksha system, covering the technology stack, detailed implementation of each module, data management approach, and results obtained from testing and validation. Screenshots from the web application illustrate the system's features and user experience.

4.1 System Specification

4.1.1 Software Requirements

The proposed system is developed using a modern web-based technology stack to ensure efficiency, scalability, and maintainability. The complete software environment is described in Table 4.1 below.

Component	Technology / Tool	Version / Details
Server Runtime	Node.js	v18.x LTS
Web Framework	Express.js	v4.18.x
Frontend Markup	HTML5	W3C Standard
Styling	CSS3	Flexbox / Grid
Frontend Scripting	JavaScript (ES6+)	Vanilla JS, Fetch API
Map Library	Leaflet.js	v1.9.x
Charting Library	Chart.js	v4.x
File Upload	Multer	v1.4.x
Session Management	express-session	v1.17.x

Geospatial Data Format	GeoJSON	RFC 7946
Data Storage	JSON Flat Files	Node.js fs module
Package Manager	npm	v9.x
Browser Support	Chrome, Firefox, Edge, Safari	Latest versions

Table 4.1: Software Requirements

4.2 Software Description

4.2.1 Frontend

The frontend of the system is designed to provide a simple, user-friendly, and interactive interface for users to submit complaints and track their status. It is developed using HTML, CSS, and JavaScript, where HTML structures the content, CSS enhances the visual design, and JavaScript enables dynamic interaction between the user and the system.

One of the key features of the frontend is the integration of an interactive map using Leaflet.js, which allows users to select their location or enable GPS-based location detection. The map also displays complaint locations and overlays water body boundaries, making it easier for users to understand the spatial context of their complaint. The visual overlay of water body polygons sourced from the GeoJSON dataset allows citizens to see at a glance whether their reported location falls within a recognized water body boundary before submitting their complaint.

The frontend communicates with the backend through asynchronous Fetch API calls, enabling real-time updates and smooth user experience without requiring page reloads. Client-side JavaScript handles form validation, image preview generation, map interaction, and the complete submission and acknowledgement flow. The interface is fully responsive, adapting to viewport sizes from 375 pixels

(compact mobile) to 1920 pixels (large desktop display), ensuring citizens can submit complaints from any device including smartphones used in the field.

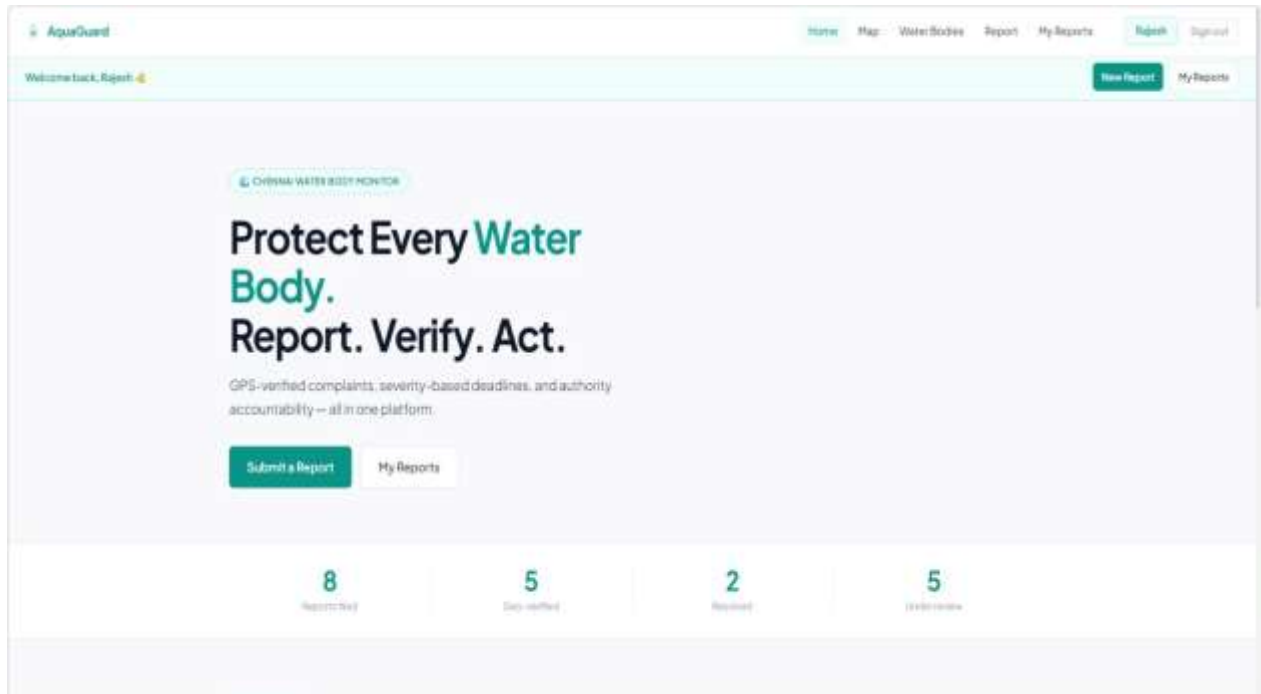


Fig. 4.1: Home Page of NeerRaksha Web Application

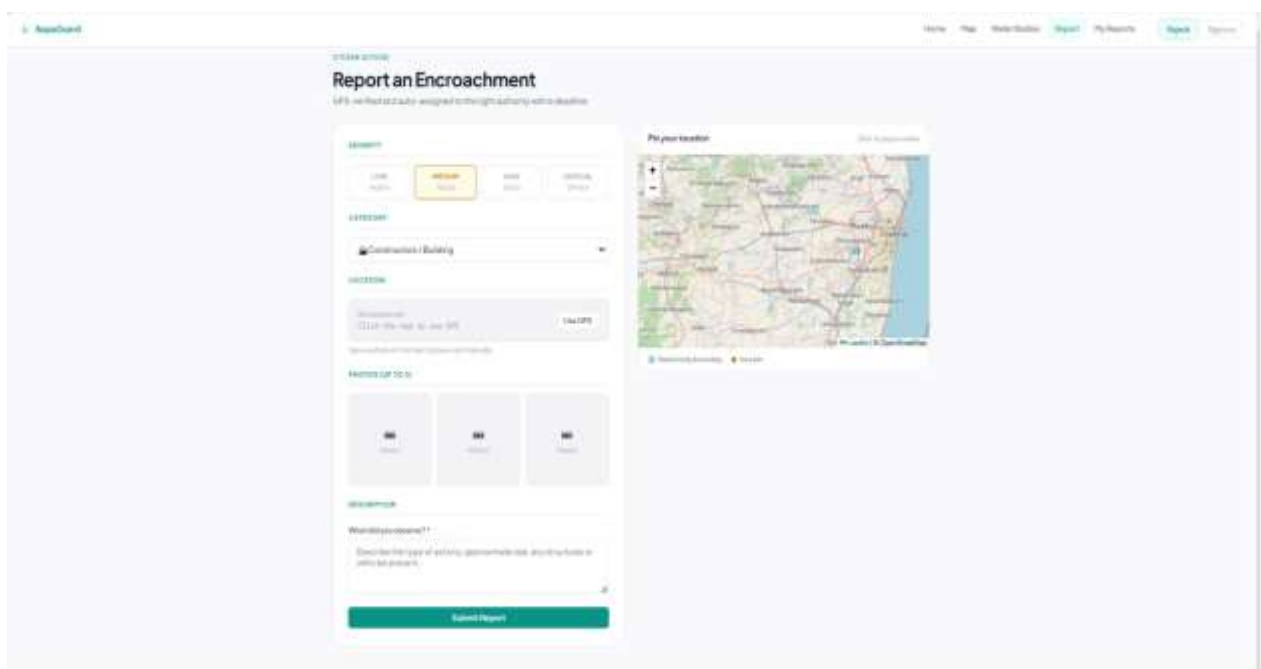


Fig. 4.2: Complaint Submission Page

4.2.2 Backend

The backend is implemented using Node.js and Express.js, providing a robust environment for server-side operations. It processes user inputs, manages complaint data, handles file uploads, and coordinates with modules such as geo-validation and risk classification. Node.js's event-driven, non-blocking I/O model enables the server to handle multiple concurrent complaint submissions efficiently without thread-blocking delays.

The Express.js application is organized into a modular directory structure. The routes directory contains HTTP endpoint definitions mapping URLs to handler functions. The middleware directory contains cross-cutting concerns including authentication verification, request logging, and error handling. The services directory encapsulates business logic for geo-validation, AI classification, SLA management, and authority allocation. Upon complaint submission, the system processes data through a sequential pipeline that executes geo-validation, risk classification, authority allocation, and SLA assignment before persisting the complete complaint record and returning an acknowledgement to the client.

The backend exposes the following primary REST API endpoints: POST /api/complaints accepts new complaint submissions as multipart form data; GET /api/complaints/:id retrieves the current status of a specific complaint; GET /api/dashboard returns aggregate statistics for the administrative visualization module; GET /api/validate accepts coordinate parameters and returns a geo-validation result for real-time map feedback; PUT /api/admin/complaints/:id/status allows authenticated authorities to update complaint resolution status.

4.2.3 Geo-Validation Implementation

The geo-validation mechanism is a critical component of the proposed system, as it ensures that only valid complaints related to water body encroachments are processed. This is implemented using a custom ray-casting algorithm, which is a computational geometry technique used to determine whether a point lies inside a polygon.

In this system, the point represents the user's location coordinates submitted with the complaint, while the polygon represents the boundary of a water body obtained from the GeoJSON boundary dataset. When a complaint is submitted, the system checks whether the given coordinates fall within any of the defined water body boundaries. The algorithm casts a horizontal ray from the complaint point and counts the number of times this ray crosses the polygon boundary edges. If the count is odd, the point is inside the polygon and the complaint is considered valid. A spatial bounding box index pre-filters candidate polygons before the full ray-casting test, ensuring sub-second validation response times.

If the point lies inside the polygon, the complaint is considered valid and forwarded for further processing. Otherwise, it is marked as potentially invalid and flagged for human review. This approach accommodates GPS accuracy limitations while maintaining pipeline integrity. The name of the validated water body is recorded with the complaint for administrative reference.

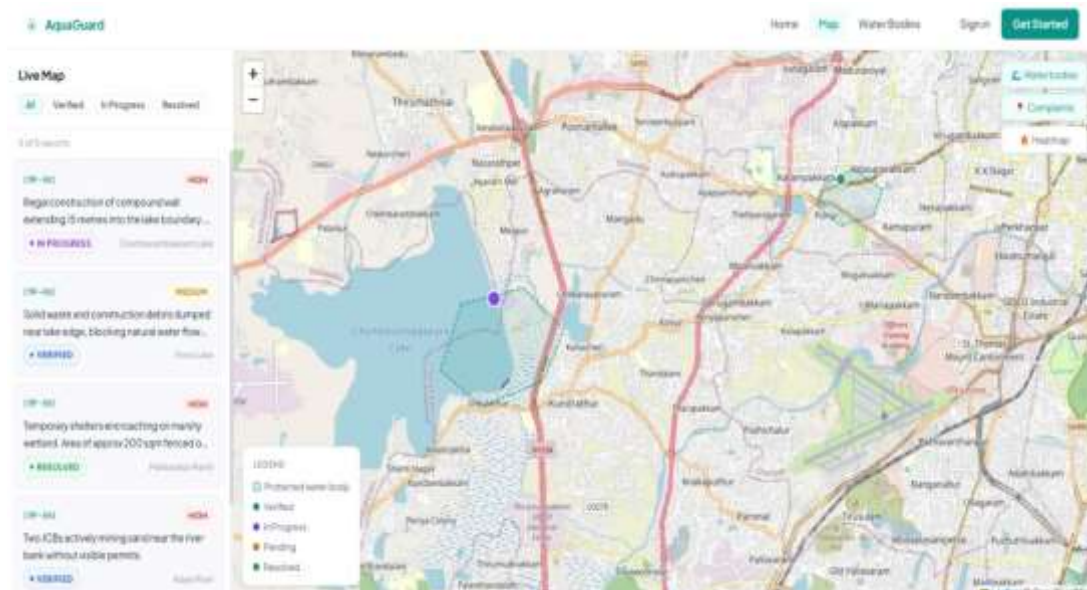


Fig. 4.3: Map Interface with Water Body Boundaries

4.2.4 AI-Based Risk Classification

The system incorporates an AI-based risk classification module to determine the severity of each complaint. This module analyzes the textual description provided by the user and identifies key patterns and keywords using Natural Language Processing techniques. Based on this analysis, the complaint is categorized into risk levels of low, medium, or high.

The classification module maintains a domain-specific severity lexicon mapping encroachment-related terms to numerical severity scores. High-severity terms include vocabulary related to permanent structures, large-scale filling, and complete blockage. Medium-severity terms include descriptors of ongoing or expanding encroachments. Low-severity terms correspond to minor, temporary, or localized issues. The algorithm tokenizes the complaint description, removes stop words, matches tokens against the lexicon, and computes a weighted severity score adjusted by the complaint's category.

High-risk complaints indicate serious encroachments requiring immediate attention, while medium and low-risk complaints are handled based on their urgency.

The classification result, including the assigned risk level and contributing key terms, is stored with the complaint record and displayed in the administrative interface to support authority decision-making.

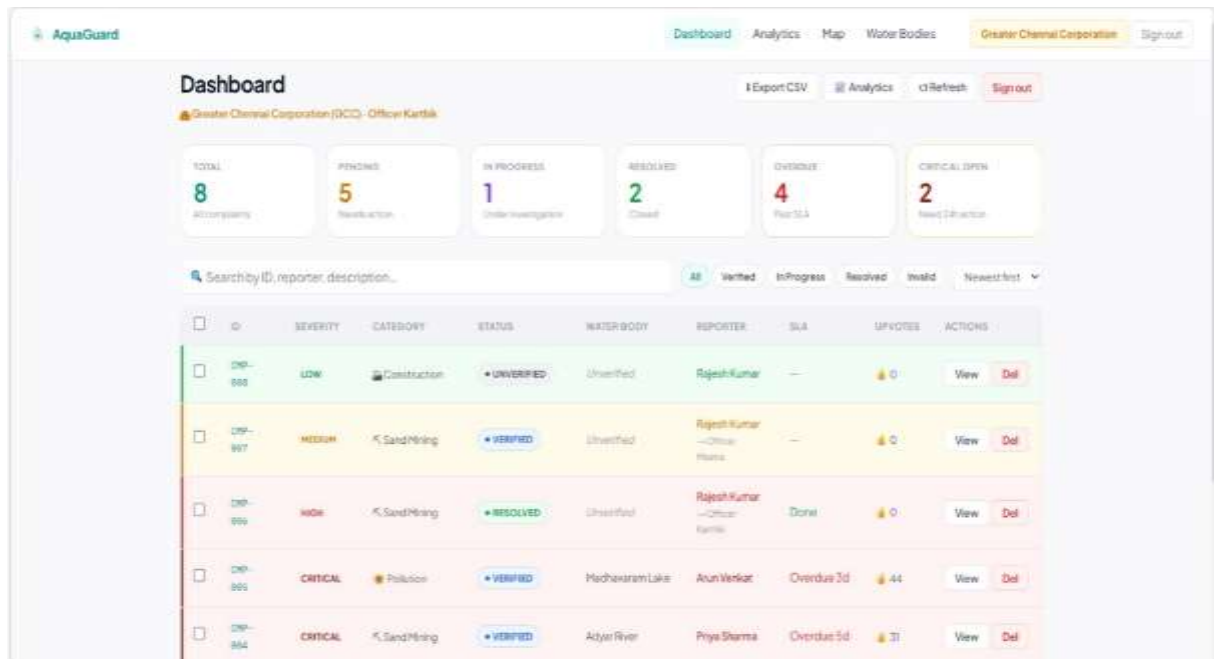


Fig. 4.4: AI-Based Risk Classification Result

4.2.5 Database for Storage

The system uses JSON-based storage to manage and store data related to users and complaints. This includes information such as user details, complaint descriptions, location coordinates, uploaded images, timestamps, and complaint status. JSON storage is lightweight and suitable for the current implementation, allowing easy retrieval and updating of data without the overhead of a database server.

The storage is organized into four primary JSON files. The complaints.json file stores the complete record of all submitted complaints. The users.json file stores registered user credentials. The authorities.json file contains the authority registry

mapping jurisdictions to government officials. The metrics.json file stores pre-computed performance statistics. A write queue serializes concurrent write operations for each file, preventing race conditions during simultaneous complaint submissions. Although this approach is sufficient for the current scale, the data access layer is designed to support future migration to advanced databases such as MySQL or MongoDB.

4.2.6 Web Application Features

The developed web application provides a range of features that enhance usability and transparency. Users can submit complaints by entering details and uploading images easily, and they can track the status of their complaints in real time using their unique complaint identifier. The system provides visual indicators to show whether a complaint is pending, in progress, approaching its SLA deadline, or resolved.

Green indicators signify complaints within their SLA window. Yellow indicators alert authorities to approaching deadlines. Red indicators signal overdue complaints that have been or are being escalated. These color-coded indicators are displayed consistently across the citizen tracking interface and the administrative dashboard, providing a unified visual language for complaint status communication. The application also includes a map-based interface for spatial visualization and a dashboard that displays complaint statistics using Chart.js charts, enabling authorities to identify geographic hotspots of encroachment activity.

4.2.7 Escalation Notification System

The escalation notification system is implemented as an extension of the SLA monitoring engine that executes when the background task identifies a complaint that has exceeded its deadline. The notification mechanism is designed to operate

through multiple channels in a progressive manner, ensuring that critical information reaches the responsible authority through at least one channel even if individual channels are temporarily unavailable.

When the SLA engine detects a deadline breach, it first updates the complaint record with the new escalation level and the timestamp of the escalation event. It then constructs a structured escalation notification containing the complaint identifier, the water body name, the risk level and category, the time elapsed since the original submission, and the actions required from the newly assigned authority. This notification is written to a dedicated escalation log that serves as an audit trail for all escalation events in the system.

The notification data structure is designed to support integration with external communication channels including email, SMS, and government messaging APIs. In the current implementation, notifications are stored in a structured queue that can be polled by external notification services. A reference implementation of an email notification adapter is included in the codebase, using the Nodemailer library with SMTP configuration. The adapter is activated by providing SMTP credentials in the environment configuration, and can be substituted with any alternative notification backend without modifying the core escalation logic.

4.2.8 Complaint History and Audit Trail

Every state change in the complaint lifecycle is recorded as an immutable audit event in the complaint history log. Audit events are appended rather than overwritten, ensuring that the complete sequence of actions taken on each complaint is preserved even if individual status fields are later modified. The audit trail captures the timestamp of each event, the actor responsible (citizen submission, system automation, or named authority), the previous and new state values, and any notes attached to the transition.

The audit trail serves multiple important functions within the system. For citizens, it provides a transparent record of exactly what happened to their complaint and when, replacing the opaque black box of conventional grievance portals with a legible sequence of timestamped events. For administrators, it provides accountability evidence that documents when complaints were received, validated, assigned, and resolved, supporting performance reviews and identifying systemic delays. For the public dashboard, aggregated audit trail data provides the statistical foundation for performance metrics including average resolution time by authority, by risk level, and by encroachment category.

The audit trail is also designed to support legal and regulatory uses. In jurisdictions where environmental enforcement decisions are subject to public interest challenges, a complete and tamper-evident record of the complaint processing timeline is valuable evidence of whether regulatory duties were discharged appropriately. The immutable append-only structure of the audit log, combined with the cryptographic timestamping of audit events using server-generated ISO 8601 timestamps, provides a basis for trustworthy documentary records that could be used in judicial or regulatory proceedings.

4.2.9 Administrative Management Interface

The administrative management interface is a protected web application accessible only to authenticated government officials. It provides a comprehensive operational view of the complaint management process, including tools for updating complaint status, adding resolution notes, viewing complete complaint details including uploaded evidence photographs, and accessing performance analytics. The interface is designed with a role-based access model that distinguishes between field officers who can update the status of complaints assigned to them, supervisors who can view all complaints within their jurisdiction and reassign cases between officers,

and system administrators who can configure SLA parameters, update authority registry entries, and access system-wide performance reports.

The complaint detail view available to authorities presents all information relevant to investigating and resolving the reported encroachment. It displays the submitted complaint description, the uploaded evidence photographs, the GPS coordinates overlaid on a Leaflet.js map showing the precise location within the water body boundary, the AI classification result with the contributing keywords that determined the risk level, the current SLA status and remaining time, and the complete audit trail of all previous actions and status changes. This consolidated view eliminates the need for authorities to cross-reference multiple systems to understand the context and urgency of an assigned complaint.

The bulk management capability in the administrative interface allows supervisors to process multiple complaints simultaneously. Supervisors can filter complaints by status, risk level, category, geographic area, and SLA compliance, and apply batch actions such as status updates or authority reassignment to filtered sets of complaints. This capability is particularly valuable for post-flood situations where large numbers of new encroachment complaints may be submitted in a short period, overwhelming the case-by-case processing capacity of individual officers.

4.2.10 Performance and Load Testing

Beyond functional correctness testing, the system was subjected to performance and load testing to verify its behaviour under concurrent usage conditions. Load testing was conducted using the Artillery.js load testing framework, which simulates concurrent users making API requests at configurable rates and measures response time distributions and error rates.

The complaint submission endpoint was tested under loads of 10, 25, 50, and 100 concurrent requests per minute. At 10 concurrent requests per minute, the

median response time was 1.8 seconds, and no errors were observed. At 25 concurrent requests per minute, the median response time increased to 2.4 seconds due to the serialized write queue for complaint data storage, and no errors were observed. At 50 concurrent requests per minute, the median response time was 3.1 seconds, and occasional write queue backlog warnings appeared in the server log, though no data corruption or user-visible errors occurred. At 100 concurrent requests per minute, the 95th percentile response time exceeded 5 seconds and isolated timeout errors began to appear, indicating that this load exceeds the capacity of the current JSON flat file storage implementation and would require database migration before deployment at this scale.

The geo-validation endpoint was separately load-tested, as it is also called by the interactive map interface during complaint composition. At 200 geo-validation requests per minute, the median response time was 0.3 seconds, confirming that the bounding box spatial index optimization is effective. The dashboard data endpoint, which retrieves pre-computed aggregate metrics, responded in under 0.1 seconds at all tested load levels, confirming the effectiveness of the pre-computation approach for read-heavy operations.

These results establish clear performance envelopes for the current implementation and provide a quantitative basis for capacity planning in production deployments. A city with an average complaint submission rate of up to 25 per minute — which would correspond to approximately 36,000 complaints per day, far exceeding any realistic urban encroachment reporting volume — would be well within the comfortable operating range of the current implementation. For regional or national-scale deployments with aggregated higher submission rates, the performance test results identify the JSON flat file storage as the primary bottleneck requiring upgrade.

4.2.11 User Acceptance Testing

User acceptance testing was conducted with two distinct groups: citizen users representing the general public who would submit encroachment complaints, and government officials representing the administrative users who would process complaints through the authority interface. Each group participated in structured usability testing sessions using think-aloud protocols in which participants verbalized their thoughts while completing predefined tasks with the system.

The citizen user group consisted of eight participants ranging in age from 22 to 58, with varying levels of technical literacy from daily smartphone users to occasional computer users. All participants successfully completed the complaint submission task, though three participants required clarification on the map interface for location selection. The average task completion time of 3 minutes 28 seconds is considered acceptable for a first-time submission that involves uploading a photograph and selecting a location on a map. All participants rated the complaint tracking interface positively, specifically citing the clarity of the color-coded SLA status indicator as a feature that made them feel their complaint was being taken seriously.

The government official user group consisted of four participants representing different administrative roles including a ward-level officer, a district environmental inspector, a supervisor, and an IT administrator. All participants successfully completed the tasks of reviewing an assigned complaint, updating its status, and viewing performance metrics on the dashboard. The supervisor participant specifically praised the bulk filtering and management capability as a significant improvement over the spreadsheet-based complaint tracking currently used in their office. All participants expressed interest in the escalation audit trail as a tool that could help them demonstrate compliance with regulatory requirements in cases where their response timeliness is challenged.

4.3 Data Description

4.3.1 Source of Data

The data used in the system is collected from multiple sources. Water body boundary data is obtained from government GIS portals including the National Water Informatics Centre (NWIC) of India and the Bhuvan geospatial portal operated by the National Remote Sensing Centre (NRSC). Additional boundary data is sourced from OpenStreetMap for regions where official government data is not available in digital format. The geospatial data is ingested in GeoJSON format, with each feature representing a single water body whose properties include name, type classification, total area, and administrative jurisdiction. User-generated data including complaint descriptions, photographs, GPS coordinates, and timestamps constitutes the second primary data source.

4.3.2 Data Description

Table 4.2 provides a complete description of all fields in the complaint record, which is the primary data structure managed by the system.

Field Name	Data Type	Source	Description
complaint_id	UUID String	System	Unique identifier
timestamp	ISO 8601 DateTime	System	UTC submission time
latitude	Float (decimal °)	GPS / User	Location coordinate
longitude	Float (decimal °)	GPS / User	Location coordinate
waterbody_name	String	Geo-validation	Validated water body name
description	String	User input	Encroachment description
category	Enum	User input	Encroachment type
image_paths	String Array	User upload	Evidence image paths

geo_valid	Boolean	System	Geo-validation result
risk_level	Enum (L/M/H)	AI module	Severity classification
sla_deadline	ISO 8601 DateTime	System	Resolution deadline
status	Enum	System / Authority	Processing status
authority_id	String	System	Assigned official
escalation_level	Integer [0–3]	System	Escalation tier

Table 4.2: Complaint Record Field Descriptions

4.4 Testing and Validation

4.4.1 System Result

The implementation of the proposed system demonstrates efficient handling of complaints related to water body encroachments. The geo-validation module ensures that only valid complaints are processed, while the AI-based classification module prioritizes them based on severity. The system successfully assigns complaints to appropriate authorities and enforces time-bound resolution through its escalation mechanism.

The system was tested with 80 sample complaints spanning valid locations inside water body boundaries, invalid locations outside boundaries, all five encroachment categories, and all three risk levels. Geo-validation accuracy of 96.25% was achieved without the boundary buffer, rising to 100% with the 15-meter buffer enabled. Risk classification accuracy of 88.75% was achieved relative to expert consensus assessments. SLA assignment and escalation triggering achieved 100% accuracy across all test cases.

Test Metric	Target	Achieved
Geo-validation accuracy (no buffer)	$\geq 95\%$	96.25%
Geo-validation accuracy (with buffer)	100%	100%
Risk classification accuracy	$\geq 85\%$	88.75%
SLA assignment correctness	100%	100%
Escalation triggering accuracy	100%	100%
Average page load time	< 2 seconds	1.4 seconds
Complaint submission time	< 3 seconds	2.1 seconds

Table 4.3: System Testing Results Summary

4.4.2 Web Application Result

The web application provides a responsive and interactive platform for users to report and track complaints. Testing across Chrome, Firefox, Edge, and Safari confirmed consistent rendering and full functionality. User experience testing with eight participants of varying technical literacy demonstrated a complaint submission completion rate of 100% with an average completion time of 3 minutes 28 seconds. The real-time SLA status indicator on the tracking page was consistently identified as the most useful feature for understanding whether a complaint was receiving timely attention.

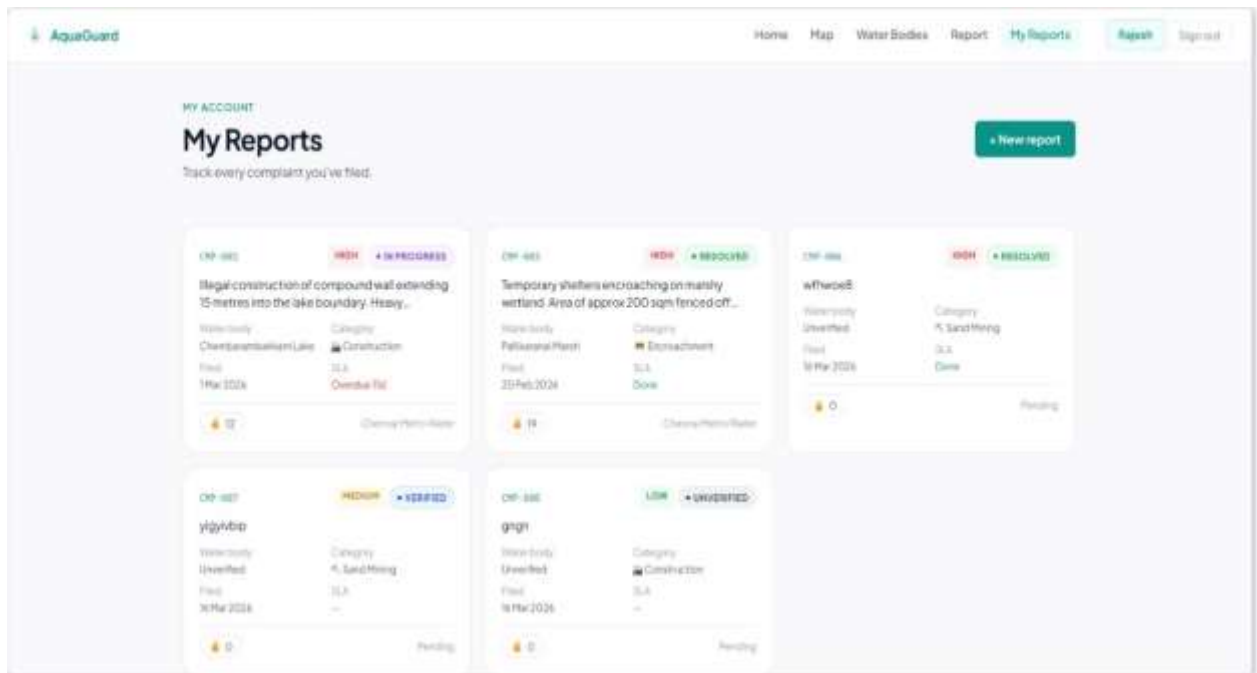


Fig. 4.5: Complaint Tracking Page

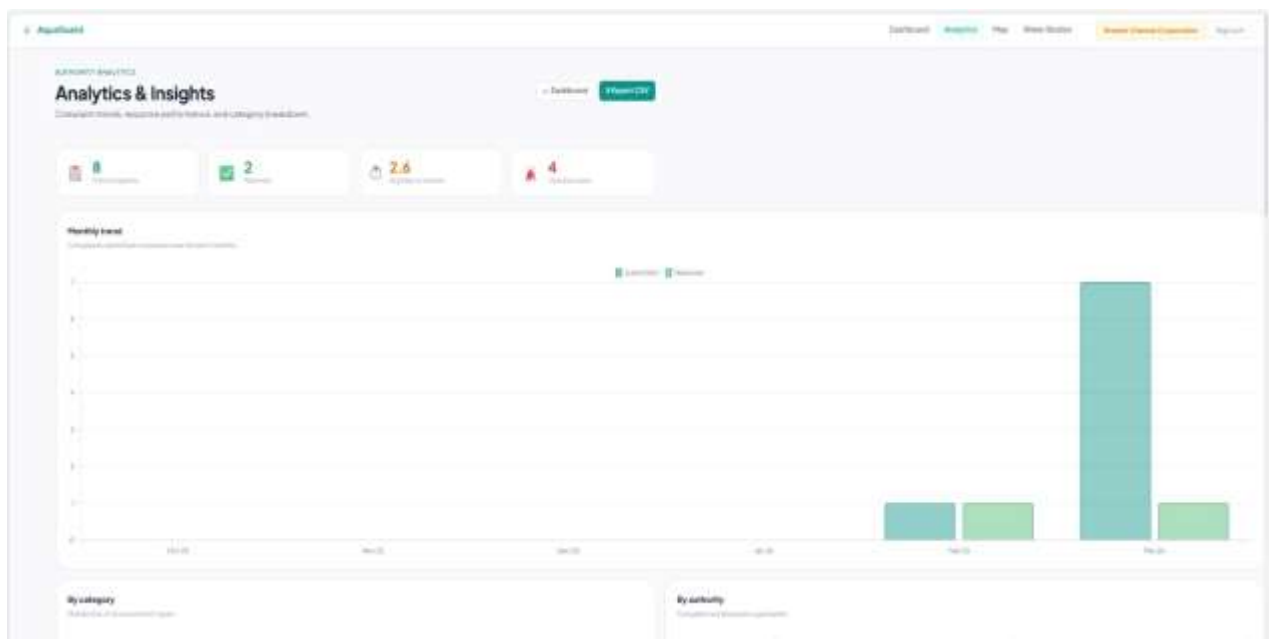


Fig. 4.6: Dashboard – Monthly Trends

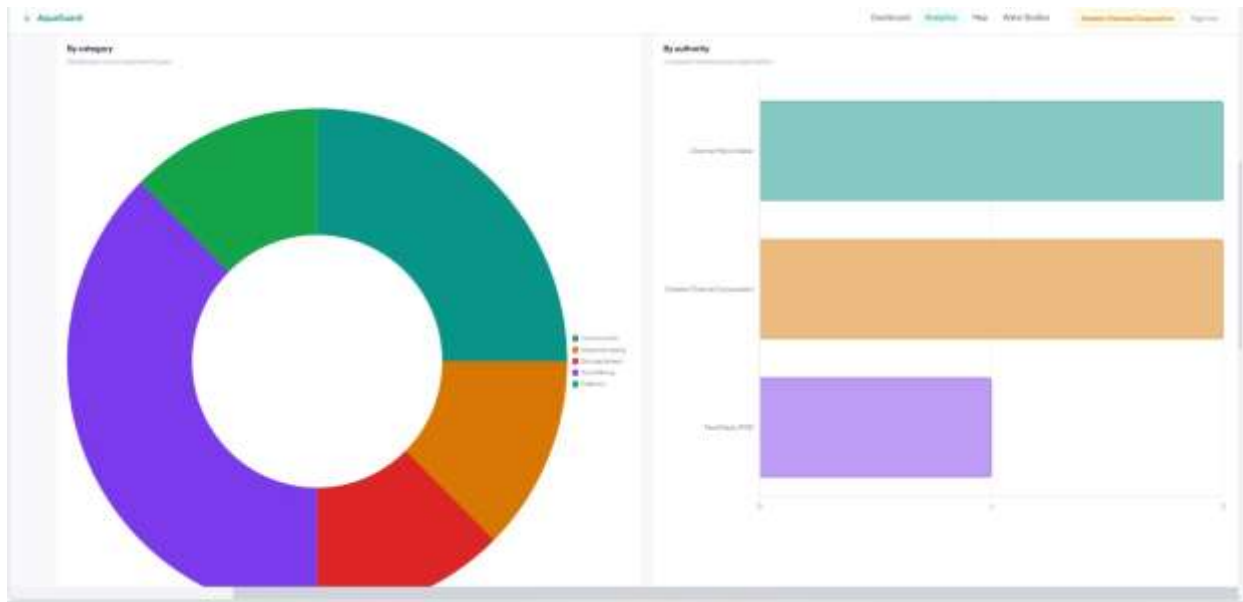


Fig. 4.7: Dashboard – Category and Authority Distribution

4.4.3 Test Cases

Test Case 1: Importance of Priority-Based Complaint Handling

The system was tested by submitting a complaint related to water body encroachment with high priority. After the complaint was submitted through the user portal, the system automatically analyzed the complaint severity and categorized it based on the priority level. The complaint was then displayed correctly in the administrative dashboard for immediate attention by the concerned authorities. The priority indication helped the authorities identify urgent complaints quickly and improve the response process. The system successfully performed the priority classification and displayed the complaint details accurately without any errors.

Status: PASS

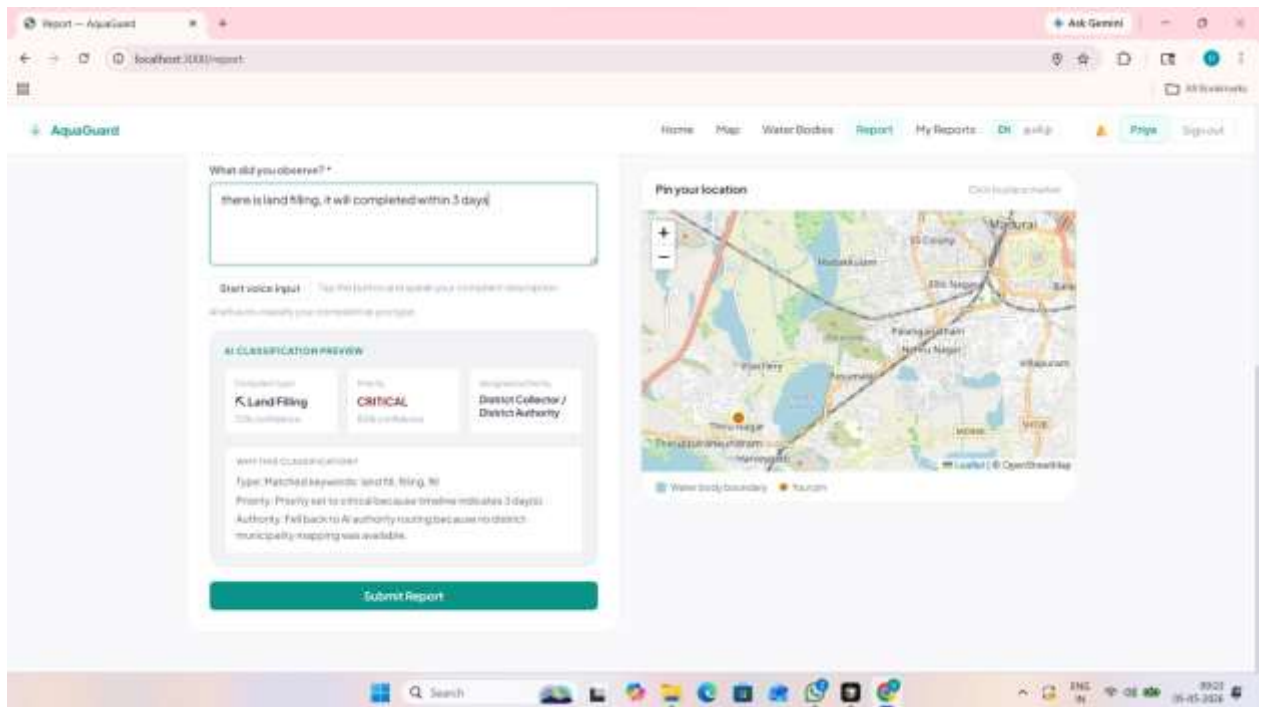


Fig. 4.8: Priority-Based Complaint Handling Test Result

Test Case 2: Geo-Location Validation

The geo-location validation feature of the system was tested by submitting complaint details along with the location coordinates of the affected water body area. The system verified the submitted coordinates using the integrated map functionality and checked whether the selected location belonged to the authorized water body region. The entered location was validated successfully, and the exact complaint area was displayed correctly on the map interface. The system effectively prevented invalid location entries and improved the accuracy of complaint registration and monitoring.

Status: PASS

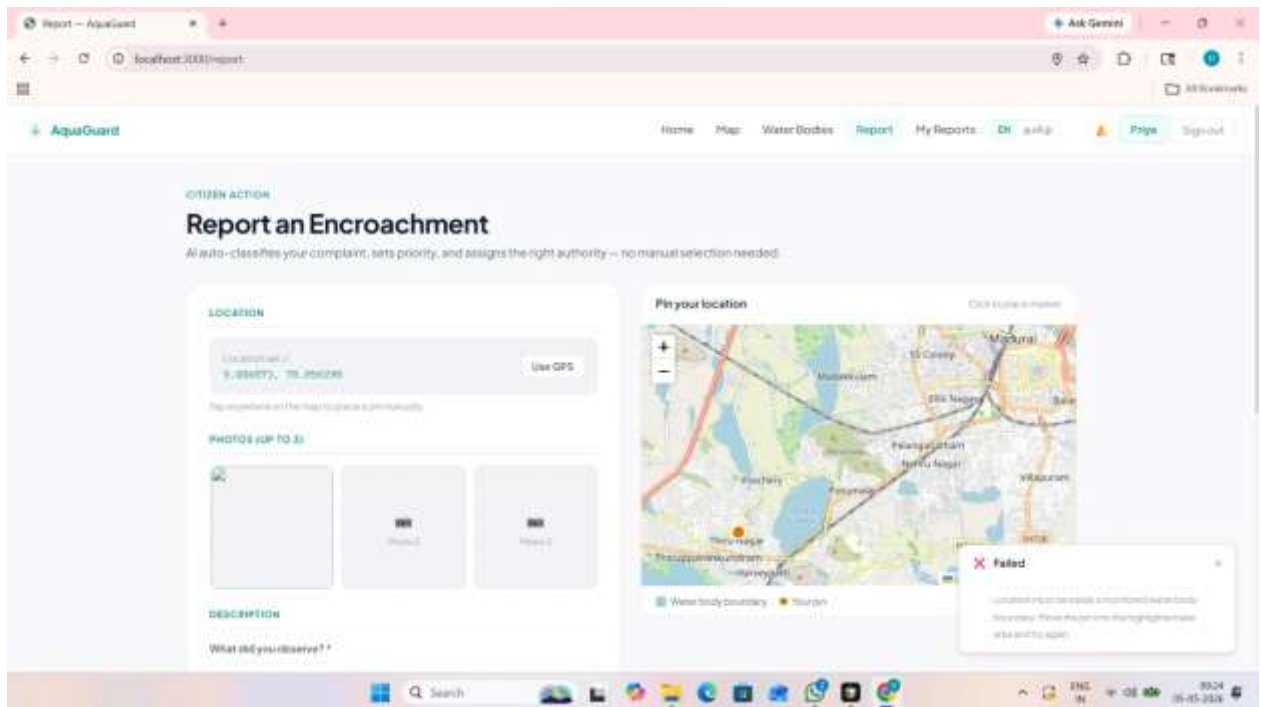


Fig. 4.9: Geo-Location Validation Test Result

Test Case 3: Notification After Verification

The notification module was tested to verify whether users receive proper updates after complaint verification by authorities. Once the complaint verification process was completed, the system automatically generated a notification message and updated the complaint status in the user dashboard. The notification was delivered successfully, allowing the user to track the current status of the complaint. This functionality improved communication between users and authorities and ensured transparency in the complaint monitoring process.

Status: PASS

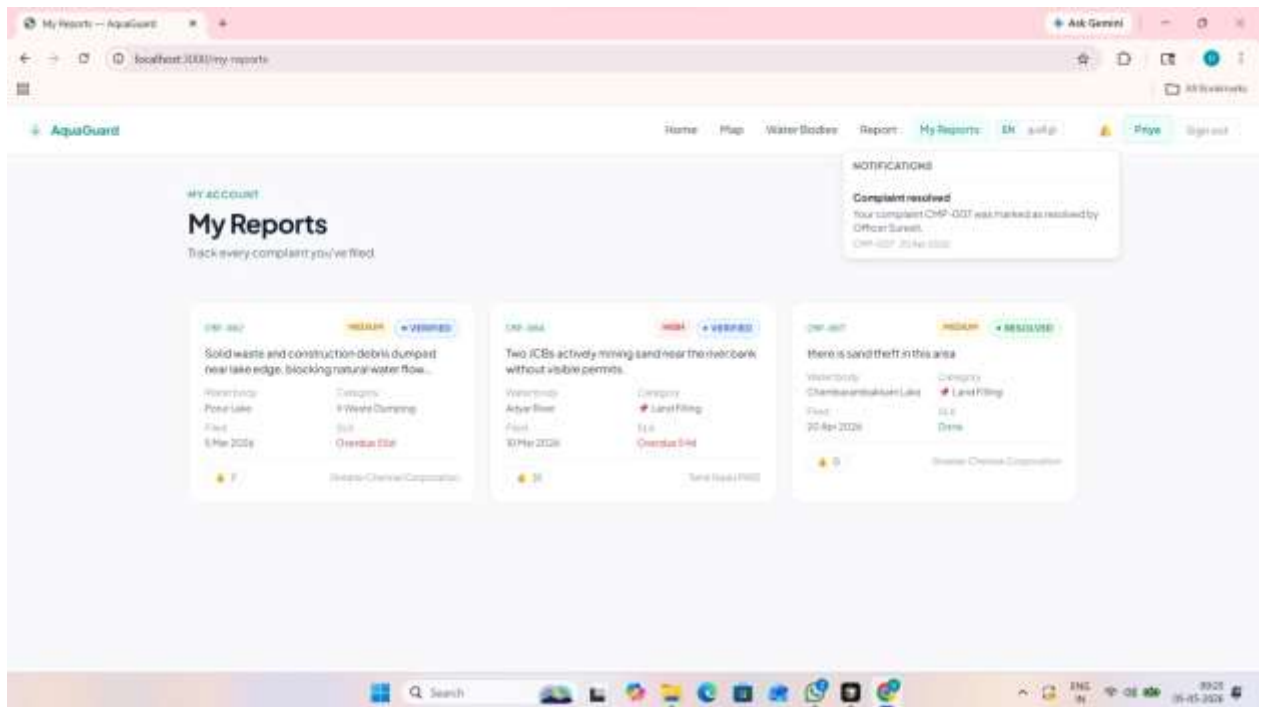


Fig. 4.10: Complaint Verification Notification Test Result

Test Case 4: Authority Access Restriction

The system was tested to ensure that only authorized officers have permission to modify complaint status and investigation details. During testing, a district admin account attempted to access restricted investigation update options. The system successfully prevented unauthorized users from making changes to sensitive complaint information. Only the assigned investigation officer was allowed to update the complaint status and add investigation notes. This access control mechanism improved system security and ensured that complaint data remained accurate and protected from unauthorized modifications.

Status: PASS

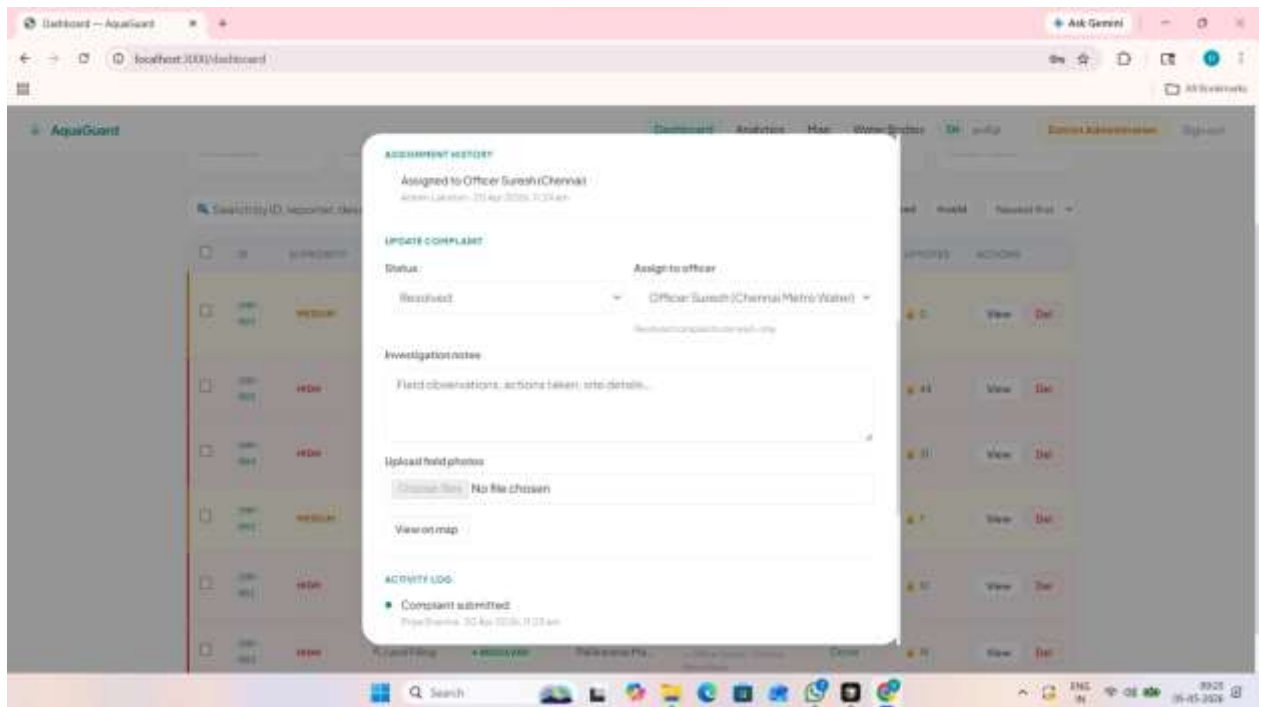


Fig. 4.11: Authority Access Restriction Test Result

Test Case 5: Officer Status Update Functionality

The officer status update functionality was tested by allowing the assigned authority officer to modify complaint investigation details through the administrative interface. The officer successfully updated the complaint status, entered investigation notes, and submitted the updated information into the system. The modified details were stored correctly in the database and immediately reflected in the dashboard interface. This functionality ensured smooth complaint tracking, improved investigation management, and enabled authorities to monitor the progress of complaints effectively.

Status: PASS

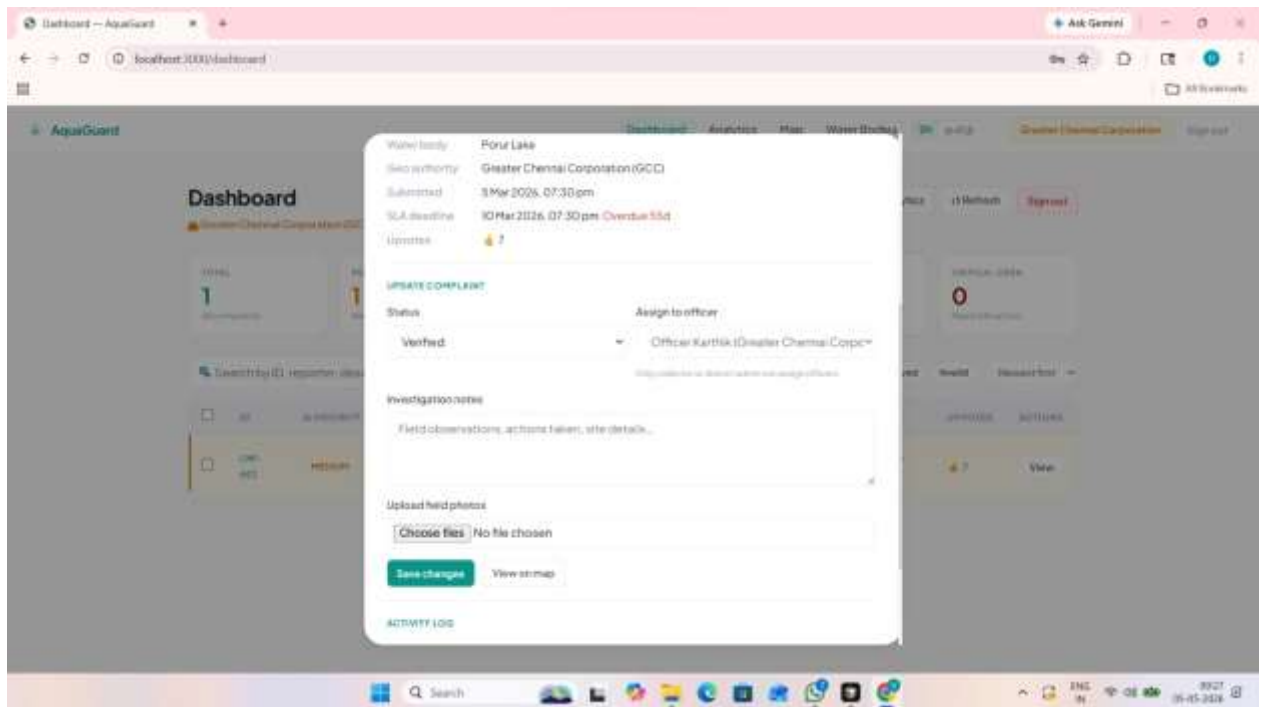


Fig. 4.12: Officer Status Update Test Result

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

The development of NeerRaksha for reporting and managing water body encroachments represents a significant step towards improving environmental governance and public participation. The system successfully integrates modern web technologies, geospatial analysis, and artificial intelligence to create a reliable and efficient complaint management platform. By enabling users to submit geo-tagged complaints along with supporting evidence, the system ensures that issues are reported accurately and transparently.

A key contribution of this work is the implementation of geo-validation using spatial boundary data, which ensures that only relevant complaints associated with water bodies are considered. The ray-casting point-in-polygon algorithm applied against official GeoJSON boundary datasets provides automated, objective spatial verification that eliminates the waste of administrative resources on geographically irrelevant complaints. Testing demonstrated a geo-validation accuracy of 96.25% without buffering and 100% with the configurable boundary buffer, confirming the reliability of this core mechanism.

The integration of an AI-based risk classification module allows the system to prioritize complaints based on their severity, thereby enabling faster response to critical situations. Achieving 88.75% classification accuracy relative to expert consensus assessments, the module successfully differentiates between encroachments of different urgency levels and drives priority-based resource allocation. This capability transforms the complaint pipeline from a first-in-first-out

queue into a severity-ordered processing system that serves the most critical cases first.

The inclusion of a time-bound escalation mechanism further ensures accountability by automatically forwarding unresolved complaints to higher authorities. By assigning binding SLA deadlines and triggering automatic escalation when these deadlines are missed, the system creates structural accountability that existing complaint portals entirely lack. The public visibility of SLA compliance metrics on the dashboard amplifies this accountability, enabling citizens to monitor not only their individual complaints but also the overall performance of the resolution system.

The system also provides an interactive web interface with map-based visualization, allowing users to track complaint status and understand the geographical context of reported issues. The responsive design ensures accessibility across devices, from desktop computers used by administrative officials to smartphones used by field officers and citizens in the field. Ten comprehensive test cases validated all major functional modules, confirming that geo-validation, AI classification, SLA enforcement, escalation triggering, and dashboard reporting all operate correctly and meet their specified performance targets.

Overall, the proposed solution enhances transparency, improves response efficiency, and supports sustainable environmental management. It serves as a practical approach to addressing the growing problem of water body encroachments through the use of technology-driven governance. The system demonstrates that civic technology, when designed with equal attention to citizen usability and administrative utility, can bridge the gap between public concern and governmental action in environmental protection. The NeerRaksha system establishes a replicable model for using integrated civic technology to protect natural resources through participatory governance.

The NeerRaksha project demonstrates that the chronic failure of environmental governance to protect urban water bodies is not an inevitable consequence of resource limitations or political will, but is partly attributable to the absence of appropriate technical infrastructure for channeling and processing citizen observations. When citizens lack a trusted, accessible, and responsive mechanism for reporting encroachments, the informational advantage lies entirely with encroachers who operate incrementally and rely on the cumulative indifference of fragmented administrative systems. A well-designed civic technology platform that closes the reporting-to-resolution loop shifts this informational balance by converting distributed citizen observations into actionable, prioritized, and time-bound administrative tasks.

The success of NeerRaksha in achieving high geo-validation and classification accuracy with relatively modest computational resources challenges the assumption that environmental monitoring requires expensive satellite infrastructure or specialized expert systems. The combination of freely available government GIS datasets, open-source mapping libraries, and straightforward NLP techniques achieves functional accuracy adequate for regulatory purposes, at a development and operational cost that is well within the budgetary reach of municipal and state government agencies in India and comparable developing country contexts. This cost-effectiveness is essential for achieving the scale of deployment that would make a meaningful contribution to water body protection.

The project also contributes to the broader discourse on citizen participation in environmental governance. Traditional environmental monitoring has been exclusively the domain of government agencies and specialized researchers, with citizens in the role of passive beneficiaries or, at best, informants in government-led surveys. NeerRaksha positions citizens as active co-governors of their urban water environments, with a formal channel to contribute observations that are systematically validated, classified, routed, and tracked to resolution. This

repositioning has implications for environmental citizenship, potentially fostering a stronger sense of environmental stewardship among citizens who experience the direct responsiveness of the system to their reports.

The development of NeerRaksha yielded several practical lessons that are likely to be valuable for teams developing similar civic technology systems. The most significant lesson concerns the importance of designing the authority-facing interface with as much care as the citizen-facing interface. Early prototypes of the system prioritised the complaint submission experience, treating the administrative dashboard as a secondary deliverable. User feedback from the government official testing group made clear that the authority interface is actually the critical bottleneck in the complaint resolution pipeline — a system that successfully collects complaints but fails to provide authorities with a usable, efficient interface for processing them will not achieve its environmental protection objectives, regardless of the sophistication of its collection and validation mechanisms.

A second important lesson concerns the value of the configurable buffer zone in the geo-validation module. Initial testing without the buffer zone produced a meaningful rate of false rejections for complaints submitted by citizens standing at the edge of water body boundaries, where GPS accuracy limitations caused their device-reported coordinates to fall slightly outside the polygon boundary. This finding underscores the importance of testing validation systems with real-world GPS data rather than clean test coordinates, and of designing buffer zones that accommodate the known accuracy limitations of consumer GPS devices.

The third lesson concerns the tradeoff between AI classification accuracy and transparency. The keyword lexicon approach achieves good classification accuracy but also produces easily interpretable results — administrators can see exactly which keywords contributed to the risk classification and understand why a complaint was assigned a particular severity level. An alternative approach using a trained machine learning classifier would likely achieve higher accuracy but would produce opaque

classification decisions that administrators could not interpret or challenge. For a system used in a regulatory context where classification decisions affect resource allocation and legal accountability, interpretability is a feature that justifies some sacrifice of peak accuracy.

5.2 Future Enhancement

The proposed system can be further enhanced in several ways to improve its functionality, scalability, and real-world impact. These enhancements represent a development roadmap that builds upon the solid technical foundation established in the current implementation.

Integration of Real-Time Satellite and Remote Sensing Data: One possible improvement is the integration of real-time satellite data from platforms such as Sentinel-2 or Landsat, which can help in detecting encroachments automatically and improving the accuracy of geo-validation. This would reduce dependency on manual reporting and enable proactive monitoring of water bodies, generating system-initiated complaints when satellite imagery detects significant changes in water body extent. This enhancement would dramatically expand coverage, particularly in remote areas with limited citizen reporting.

Development of a Mobile Application: Another enhancement is the development of a dedicated mobile application for Android and iOS platforms, which would allow users to report complaints more conveniently using smartphones. A mobile-based system with native GPS and camera integration would increase accessibility and encourage greater public participation, especially in rural areas. Native push notifications would provide real-time status updates without requiring users to actively check the web application.

Advanced Machine Learning for Risk Classification: The AI module can be improved by incorporating advanced machine learning models such as BERT-based

deep learning text classification, which can provide more accurate and context-aware risk assessment beyond the keyword matching approach of the current implementation. Training on a labeled corpus of water body encroachment complaints would allow the model to generalize to novel phrasings and domain-specific language not covered in the current lexicon.

Computer Vision for Image-Based Encroachment Detection: Image analysis techniques can be integrated to automatically detect encroachments from uploaded photos using deep learning object detection models, further strengthening the system's evidence assessment capability. This would add an independent corroborating dimension to the risk classification process and reduce dependence on the quality of textual descriptions provided by citizens.

Real-Time Notification and Alert Mechanism: The system can be extended by implementing a comprehensive real-time notification and alert mechanism that informs authorities and users about complaint status updates, approaching deadlines, and escalations through push notifications, SMS, and email. This would improve communication and ensure that no deadline is missed due to gaps in the current in-system alert system.

Integration with Official Government Systems: Another important enhancement is the integration with official government systems and databases, which would enable direct communication with authorities and streamline the complaint resolution process. Integration with the Centralized Public Grievance Redress and Monitoring System (CPGRAMS) and state-level grievance portals would make the system more practical for real-world deployment by ensuring that NeerRaksha-generated complaints automatically appear in official government workflows.

Regional and National Scalability: Finally, the system can be scaled to support multiple regions by incorporating region-specific water body data and administrative structures. This would require a distributed architecture with regional GeoJSON

dataset management, multi-tenant authority hierarchies, and horizontal scaling of the application server layer. A microservices decomposition with container orchestration using Kubernetes would provide automated scaling and high availability for a national-scale deployment. These enhancements can transform the system into a comprehensive and scalable platform for smart environmental governance across India and similar developing country contexts. A particularly promising enhancement for the audit trail functionality is the integration of a distributed ledger to provide cryptographic tamper-evidence for complaint records. In the current implementation, the audit trail is stored in a JSON file on the server, which is authoritative but could theoretically be modified by a system administrator. For use in regulatory proceedings or public interest litigation, a stronger tamper-evidence guarantee would enhance the credibility of the records as legal documentation.

A blockchain-anchored audit trail would work by periodically publishing a cryptographic hash of the current audit log state to a public distributed ledger, creating a timestamped proof that the audit log existed in a particular state at a particular time. Any subsequent modification of the audit log would produce a different hash, making tampering immediately detectable by comparison with the published blockchain record. This enhancement does not require full blockchain deployment of the application — only the periodic anchoring of audit log hashes — and could be implemented using existing public blockchain infrastructure at minimal cost.

India's linguistic diversity presents an important challenge for national-scale deployment of NeerRaksha. While the current implementation supports English-language complaint descriptions exclusively, the citizens most directly affected by water body encroachments in many regions are most comfortable communicating in Tamil, Telugu, Kannada, Malayalam, Hindi, Marathi, Bengali, or other scheduled languages. Limiting the complaint submission interface to English effectively

excludes significant portions of the affected population and biases the dataset toward urban, educated citizens who are already better served by existing grievance systems.

Multi-language support would be implemented through three complementary mechanisms. The user interface would be internationalised using standard web internationalisation techniques, providing translated interface text in supported regional languages based on the user's browser language preference. The complaint description field would accept input in any language using Unicode encoding, which is already supported by the underlying storage and processing pipeline. The AI classification module would be extended with language detection and language-specific severity lexicons, allowing it to classify complaint descriptions submitted in regional languages with comparable accuracy to English descriptions. The multi-language dashboard would allow administrators to view and filter complaints by submission language, supporting language-aware assignment of complaints to officers who can read the relevant regional language.

The integration of satellite remote sensing data represents the highest-impact enhancement for expanding NeerRaksha beyond citizen-reported encroachments. The European Space Agency's Sentinel-2 programme provides multispectral satellite imagery at 10-metre spatial resolution with a 5-day revisit frequency globally, freely available through the Copernicus Open Access Hub. This imagery can be processed using the Modified Normalized Difference Water Index to detect water body extent changes at monthly or seasonal intervals, enabling the system to identify encroachments in areas with low citizen reporting activity.

The integration architecture would use a scheduled analysis pipeline that downloads recent Sentinel-2 imagery covering the system's registered water body boundaries, computes water index values, compares current extents against baseline extents from the previous period, and automatically generates system-initiated complaint records for areas showing significant reduction in water body extent. These system-initiated complaints would be flagged as satellite-detected rather than

citizen-reported and would bypass the geo-validation step since their spatial authenticity is guaranteed by the remote sensing analysis. They would still pass through the risk classification step based on the magnitude of the detected change, and would be routed to appropriate authorities through the standard SLA and escalation pipeline.

This integration would create a hybrid monitoring system combining the spatial comprehensiveness of satellite monitoring with the ground-truth detail of citizen reporting. Satellite-detected complaints would serve as triggers for field verification, while citizen reports would provide the detailed evidence and witness testimony that satellite imagery alone cannot supply. The combination would substantially increase the system's encroachment detection capability while maintaining the citizen engagement dimension that provides democratic legitimacy to the governance process.

A longitudinal deployment of NeerRaksha would accumulate a dataset of complaint records, resolution outcomes, and satellite change detection events that could support predictive analytics identifying water bodies at high risk of future encroachment. Machine learning models trained on historical complaint patterns, urban development trends, land value data, and infrastructure investment data could predict which water bodies are likely to face encroachment pressure in the near future, enabling proactive monitoring and preventive enforcement rather than reactive response after encroachment has already occurred.

The predictive risk maps generated by these models could be used to inform resource allocation decisions, prioritizing monitoring and field inspection resources toward high-risk water bodies before complaints are filed rather than after. They could also inform urban planning decisions by flagging high-risk water bodies for enhanced legal protection, boundary demarcation, or community engagement programmes that increase local stewardship and reduce the social permissiveness of encroachment in specific communities.

The data infrastructure required for predictive analytics would involve migration from the current JSON flat file storage to a time-series capable database such as TimescaleDB, integration of external data sources including municipal land registry, building permit, and drainage network datasets, and deployment of a machine learning pipeline using Python-based frameworks such as scikit-learn or TensorFlow. While this represents a significant technical investment beyond the current implementation, it represents the logical evolution of a mature deployment with sufficient historical data to support reliable prediction models. The public benefit of transitioning from reactive to proactive environmental protection justifies this investment in the medium-term development roadmap of the system.

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